

Redistricting and Municipal Finance

Repository Documentation and Replication Overview

Thesis Information

Title: Economic Effects of Splitting Cities When Redistricting: Effects of Gerrymandering Beyond Politics

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Overview

This repository contains the data, code, geographic workflows, documentation, and supporting materials for the honors thesis *Economic Effects of Splitting Cities When Redistricting: Effects of Gerrymandering Beyond Politics*.

The project examines whether splitting a municipality across multiple congressional districts affects the amount of federal funding that municipality receives. Congressional redistricting is usually studied as a political process involving elections, representation, communities of interest, and partisan competition. This thesis studies redistricting as an economic process as well, asking whether the placement of district lines has measurable fiscal consequences for local governments.

The repository serves as a replication archive for the thesis. It documents the construction of a novel national municipal panel dataset, the geographic workflows used to identify congressional district fragmentation, the empirical analysis, and the supporting materials needed to reproduce or extend the project.

Research Question

Do municipalities that are divided across multiple congressional districts experience different fiscal outcomes than municipalities represented by a single congressional district?

More specifically, does congressional district fragmentation reduce the amount of federal intergovernmental funding received by local governments?

Motivation

Congressional district boundaries determine more than electoral representation. They also shape how municipalities interact with the federal government.

A municipality entirely contained within one congressional district has a single representative who receives the full political benefit of advocating for that municipality. A municipality split across multiple congressional districts may face a collective action problem: each representative bears the cost of advocacy but captures only a portion of the electoral return.

The theoretical effect is ambiguous. Split municipalities may lose out because no single representative has full responsibility for advocating on their behalf. At the same time, they may benefit from having multiple representatives who can advocate through different committees, coalitions, and political networks.

This thesis tests which effect dominates empirically.

The project originated from a practical policy question about Fall River, Massachusetts, which was split across two congressional districts after the 2010 redistricting cycle and reunified after the 2020 cycle. That case motivated a broader national analysis of whether city splitting has measurable fiscal consequences.

Main Findings

The thesis finds evidence that congressional district fragmentation reduces federal funding received by municipalities.

Key findings include:

- Municipalities contained entirely within a single congressional district receive approximately \$3.77 more in federal intergovernmental transfers per capita each year than municipalities split across multiple districts, among municipalities receiving any federal transfers.
- Over a decade, this gap amounts to roughly \$37.65 per resident.
- For a municipality with 100,000 residents, this corresponds to approximately \$3.8 million in federal resources over a single redistricting cycle.
- Split municipalities receive between 11% and 19% fewer federal intergovernmental transfers per capita than unified municipalities across the main Census of Governments waves examined in the thesis.
- Local tax revenue is largely unaffected, suggesting that the mechanism operates through federal representation rather than broader local economic decline.

- The effect is concentrated among small and medium municipalities, where representatives serving split municipalities often also represent many other towns, intensifying the collective action problem.
- Partisan alignment matters: the fiscal penalty of fragmentation is amplified in some settings where the governor, president, and dominant congressional district are co-partisan, consistent with “taken-for-granted” dynamics in the distributive politics literature.

Taken together, the results suggest that splitting municipalities across congressional districts is not merely a representational or aesthetic concern. It may also impose measurable fiscal costs on local governments and residents.

Dataset Construction

A central contribution of this project is the creation of a new national municipal panel dataset linking congressional representation to municipal fiscal outcomes.

The dataset combines fiscal, geographic, demographic, and political data for municipal governments in the 48 contiguous states. The panel spans 1987 to 2022 and draws on nearly 100 independent data sources across state, local, federal, and political data providers.

The data construction process required substantial technical work. The thesis describes the dataset as requiring separate data acquisition, format conversion, geographic projection, and crosswalking to consistent place identifiers across Census of Governments waves. More than 8,000 lines of code were used to assemble the final research infrastructure.

The construction process began with the geographic side of the project. Individual state-level municipal boundary datasets were collected, cleaned, standardized, and merged into a national dataset. This process was difficult because state files used inconsistent formats, naming conventions, projection systems, and identifier schemes. Where possible, files were standardized to include consistent county identifiers, municipal identifiers, state FIPS codes, place-level GEOIDs, and variable names.

After municipal boundaries were standardized, they were linked to congressional district boundaries using spatial overlay methods. The geographic workflow made it possible to identify whether a municipality was entirely contained within one congressional district or split across multiple congressional districts in a given redistricting cycle.

The full data construction process involved:

- Collecting municipal boundary data from state GIS repositories, Census TIGER/Line files, and Census County Subdivision files
- Cleaning and standardizing state-level municipal boundary files
- Harmonizing geographic identifiers, including state, county, place, and municipality identifiers where available
- Projecting geographic files into common coordinate systems
- Combining state-level municipal files into a national municipal boundary dataset
- Overlaying municipal boundaries with congressional district boundaries
- Constructing municipality-to-congressional-district geographic crosswalks
- Identifying whether municipalities were unified or split across congressional districts
- Constructing a dominant district share for each municipality
- Linking geographic split status to Census of Governments fiscal data
- Merging 1990 Decennial Census demographic controls
- Incorporating Cook Partisan Voting Index data
- Constructing governor, president, and district partisan alignment variables
- Creating a municipality-level panel suitable for fixed-effects and event-study analysis

The geographic overlay and matching process is a major part of the project's research contribution.

Data Sources

The analysis combines several categories of data: fiscal data, geographic data, demographic data, and political data.

Fiscal Data

The fiscal data come from the U.S. Census of Governments.

The Census of Governments provides the institutional and fiscal foundation of the dataset. It is conducted by the U.S. Census Bureau every five years in years ending in 2 and 7. The thesis uses Census of Governments data from 1987 through 2022.

The 2007 wave is excluded because it was not released for public use.

The Census of Governments identifies legally distinct government units and collects detailed information on local government finances, employment, and structure. The project focuses on municipal governments, including cities, towns, townships, villages, and analogous general-purpose local governments depending on state law.

The primary fiscal outcome is federal intergovernmental revenue per capita. The secondary outcome is local tax revenue per capita.

Geographic Data

The geographic data come from state GIS boundary files, U.S. Census TIGER/Line files, Census County Subdivision files, and congressional district boundary files.

The baseline municipal boundary dataset primarily draws on state-maintained GIS boundary files, which provide detailed and accurate representations of municipal boundaries. State-sourced boundary data are used in 35 states.

Where state-specific municipal boundaries were not publicly available, or where Census place boundaries did not correspond well to Census of Governments municipal units, the project uses Census TIGER/Line place files or Census County Subdivision files.

This distinction is especially important in New England and parts of the Midwest, where towns, civil divisions, or townships often function as the relevant municipal government units.

Congressional district boundaries are drawn from the standardized digital boundary definitions compiled by Lewis et al. (2013), available through the UCLA congressional district boundary project. These files provide congressional district shapefiles across historical Congresses and are used to construct municipality-to-district linkages.

Demographic Data

Baseline demographic and socioeconomic controls are drawn from the 1990 Decennial Census, using the National Historical Geographic Information System compilation.

These controls include racial and ethnic composition, age and gender composition, veteran status, disability status, citizenship status, language spoken at home, years of residence in the county, median household income, household size, educational attainment, poverty rates, and selected interaction terms.

Demographic controls are fixed at their 1990 baseline values to avoid simultaneity problems. Population is treated separately and updated across Census of Governments waves using decennial Census and intercensal estimates.

Political Data

The Cook Partisan Voting Index is used to measure congressional district partisan lean. The thesis uses Cook PVI data from 1997 through 2025, covering the 105th Congress through the current 119th Congress.

The Cook PVI data are used for internal analytical purposes and are not redistributed as a standalone product.

Governor partisanship is drawn from an ICPSR panel of U.S. governors and extended through 2025 using publicly available information. Presidential partisanship is recorded for each year in the sample.

These data are used to construct partisan alignment measures between congressional districts, governors, and presidents.

Measuring Congressional Fragmentation

The thesis uses two primary measures of congressional district fragmentation: a binary split indicator and a dominant district share.

Binary Split Indicator

The binary split indicator equals one if a municipality is geographically split across more than one congressional district in a given period and zero otherwise.

A municipality is classified as unified if its entire geographic area falls within a single congressional district.

A 5% threshold is used to avoid classifying very small overlaps caused by GIS imprecision as meaningful splits.

Dominant District Share

The dominant district share is a continuous measure of fragmentation. It measures the share of a municipality located in its largest congressional district.

A higher dominant district share means the municipality is more geographically unified. A lower dominant district share means the municipality is more fragmented across districts.

The thesis treats dominant district share as the preferred continuous measure because it captures politically meaningful variation in how concentrated or fragmented a municipality's congressional representation is.

Empirical Strategy

The primary empirical strategy is a two-way fixed-effects event study.

The analysis exploits decennial redistricting as a plausibly exogenous source of variation in municipal split status. It compares municipalities that transition between unified and split status against municipalities whose representational status remains stable.

The main regression framework includes municipality fixed effects, time fixed effects, baseline demographic controls, updated population controls, district partisan composition, Cook PVI, president-district alignment, governor-district alignment, and other partisan alignment measures.

The two main outcome variables are federal intergovernmental transfers per capita and local tax revenue per capita.

Outcomes are analyzed in logs to facilitate interpretation as percentage effects.

The empirical design focuses primarily on the 1990, 2000, and 2010 redistricting cycles. The 1980 cycle lacks pre-period fiscal data, and the effects of the 2020 cycle had not fully materialized by the time of the final observed Census of Governments wave.

Robustness and Validation

The thesis includes several checks to evaluate the credibility and stability of the results. These include:

- Pre-period parallel trends tests
- A matching algorithm check
- A relative time event study
- Sensitivity to early wave exclusion
- Specifications using both the binary split indicator and dominant district share
- Separate analysis of federal transfers and local tax revenue
- Heterogeneity analysis by city size
- Heterogeneity analysis by partisan alignment

The federal transfer results are robust across multiple specifications. The local tax revenue results are generally null, supporting the interpretation that congressional district fragmentation affects federal fiscal transfers rather than broader municipal economic capacity.

Repository Structure

The repository is organized into folders for data, code, figures, presentation materials, thesis materials, and documentation.

redistricting-municipal-finance/

```
|— data/
|   |— processed/
|   |— crosswalks/
|   |— documentation/
```

```
|— code/
|   |— cleaning/
```

```
| |— gis/
| |— analysis/
| |— visualization/
```

```
|— figures/
|— presentation/
|— thesis/
|— README.md
|— LICENSE
```

Data Availability

Most data sources used in this project are publicly available and can be obtained directly from their original providers.

Some data sources are proprietary, restricted, or subject to third-party licensing limitations. In particular, Cook Partisan Voting Index data are used for internal analytical purposes and are not redistributed as a standalone dataset.

Where redistribution restrictions apply, this repository provides documentation describing the source, its role in the analysis, and the steps required for users to obtain the data independently. Large geographic files may also need to be downloaded separately from their original providers.

Replication

This repository is intended to support replication and extension of the thesis.

Replication materials include or will include:

- Data cleaning scripts
- Geographic processing workflows
- Municipality boundary standardization procedures
- Municipality-to-congressional-district crosswalk construction code
- Census of Governments fiscal data processing code
- Demographic data merging scripts
- Political data merging scripts
- Statistical analysis code
- Figure generation scripts
- Documentation of data sources and construction decisions
- Final thesis materials

Because the project relies on large geographic files and some restricted third-party data sources, full replication may require users to download certain inputs independently.

Software and Tools

The thesis identifies Python and VS Code as central tools for data processing and analysis. The project also relies on GIS and spatial overlay workflows for the construction of municipality-to-congressional-district linkages.

The repository may include workflows involving Python, VS Code, GIS shapefiles, spatial intersection methods, Census of Governments data files, geographic crosswalk construction, statistical analysis, and visualization scripts.

Citation

If you use this repository, dataset, or code, please cite:

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